

Homework 1 FAQ

Last Updated: 1/28/2017

b. number of orders

When the `ORDER_ID = 0` it means the order is non-displayable (hidden). Why it is hidden we are less sure about - possibly dark pools. Where the `ORDER_ID` equals zero you will notice that the message type is `P` which means it is a trade so you do NOT want to filter them out.

Since we do not know whether some of these trades are part of one order or all individual orders as far as the number of orders are concerned we can just subtract one (which represents all the zero order ids) from the total number of orders. Then one can report the number of non-hidden orders.

The relevant section to read more about this is in the NASDAQ TotalView-ITCH 4.1 document Section 4.6.1 which is posted in OneTick documentation.

d. vwap

Can do entirely in OneTick. See `OneTickGuiTutorial.pdf`

e. BBO spread and depth

You will need to use PRL data on this problem utilizing one of the `OB_` event processors to obtain the information you want. No need to try to reconstruct the order book on your own. Use the smallest `BUCKET_INTERVAL` possible when using your `OB_` event processor. 0 represents bid and 1 represents offer in PRL.

g. 5-second price impact

Should end up with 3 columns: midpoint quote before the trade, midpoint quote 5 seconds later and either a +1 or -1 for the order type (likely have to transform from 0 and 1 for sell and buy), you take the difference between the 2 midpoint quotes to get the price impact and can regress that on the -1 or +1